CSCI218: Foundations of Artificial Intelligence

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Outline

- Basic concepts of knowledge and logic
- Propositional logic: syntax and semantics
- Inference by model checking
- Inference by theorem proving

Agents that know things

- Agents acquire **knowledge** through perception, learning, language
  - Knowledge of the effects of actions (“transition model”)
  - Knowledge of how the world affects sensors (“sensor model”)
  - Knowledge of the current state of the world
- Can keep track of a partially observable world
- Can formulate plans to achieve goals
- A knowledge-based agent
  - Its central component is a knowledge base (KB)

Knowledge-based agent

- Knowledge base = a set of **sentences** in a formal **language**
  - Here “sentence” is used as a technical term
  - Every sentence must be either true or false (no “in between”)
  - Language: a knowledge representation language
  - A sentence represents some assertion about the world
- Axiom
  - A sentence is taken as being given without being derived from other sentences

Knowledge-based agent, contd.

- How to add new sentences to a KB?
  - ***Tell*** it what it needs to know (or have it ***Learn*** the knowledge)
- How to query a KB on what is known?
  - ***Ask*** itself what to do—answers should follow from the KB
- **Inference**: deriving new sentences from old.
- Agents can be viewed at the ***knowledge level***
  i.e., what they ***know***, regardless of how implemented
- A single inference algorithm can answer any answerable question

Knowledge base	Domain-specific facts
Inference engine	Generic code

Knowledge-based agent, contd.

```
function KB-AGENT(percept) returns an action
  persistent: KB, a knowledge base
              t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action ← ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t ← t + 1
  return action
```

A generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action.

Logic

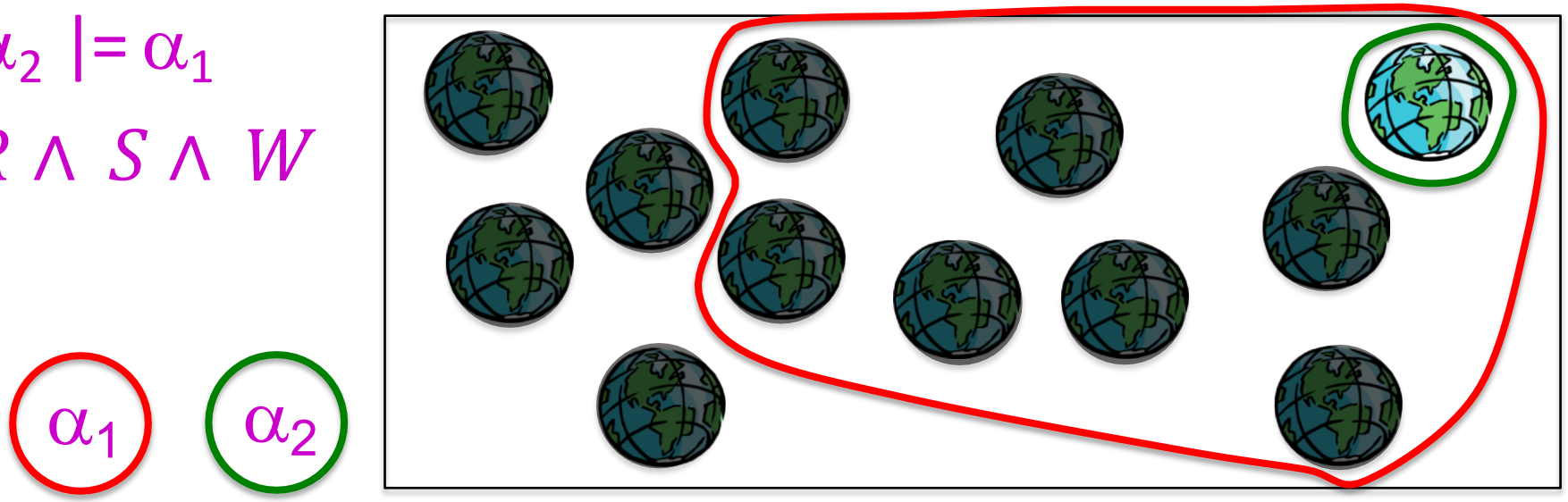
- ***Syntax***: What sentences are allowed in a knowledge base?
  - “x + y = 4” vs. “x4y+=“
- ***Semantics***: meaning of a sentence
  - It defines the truth of each sentence w.r.t. each possible world
  - When ”x + y = 4” is true?
- Model: an assignment or configuration of the world
  - “x = 2 and y = 2” vs. “x = 1 and y = 1”
  - If a sentence α is true in model *m*, we say “*m* satisfies α” or “*m* is a model of α”
  - Models(α) or $M(\alpha)$ denotes the set of all models of α

Different kinds of logic

- Propositional logic
  - Syntax: $P \vee (\neg Q \wedge R)$; $X_1 \Leftrightarrow (Raining \Rightarrow \neg Sunny)$
  - Possible world: {P=true, Q=true, R=false, S=true} or 1101
  - Semantics: $\alpha \wedge \beta$ is true in a world iff is α true and β is true (etc.)
- First-order logic
  - Syntax: $\forall x \ \exists y \ P(x,y) \wedge \neg Q\big(Joe, f(x)\big) \Rightarrow f(x) = f(y)$
  - Possible world: Objects o_1, o_2, o_3; P holds for $<o_1,o_2>$; Q holds for $<o_3>$; $f(o_1)=o_1$; Joe=o_3; etc.
  - Semantics: $\phi(\sigma)$ is true in a world if $\sigma = o_j$ and ϕ holds for o_j; etc.

Inference: entailment

- Now we are ready to talk about logic reasoning
- ***Entailment***: $\alpha \models \beta$ ("α entails β" or "β follows from α") iff in every world where α is true, β is also true
  - I.e., the α-worlds are a subset of the β-worlds [$\boldsymbol{models}(\alpha) \subseteq \boldsymbol{models}(\beta)$]
  - I.e., α is a stronger assertion than β
- In the example, $\alpha_2 \models \alpha_1$
- (Say α_2 is $\neg Q \wedge R \wedge S \wedge W$
  α_1 is $\neg Q$)



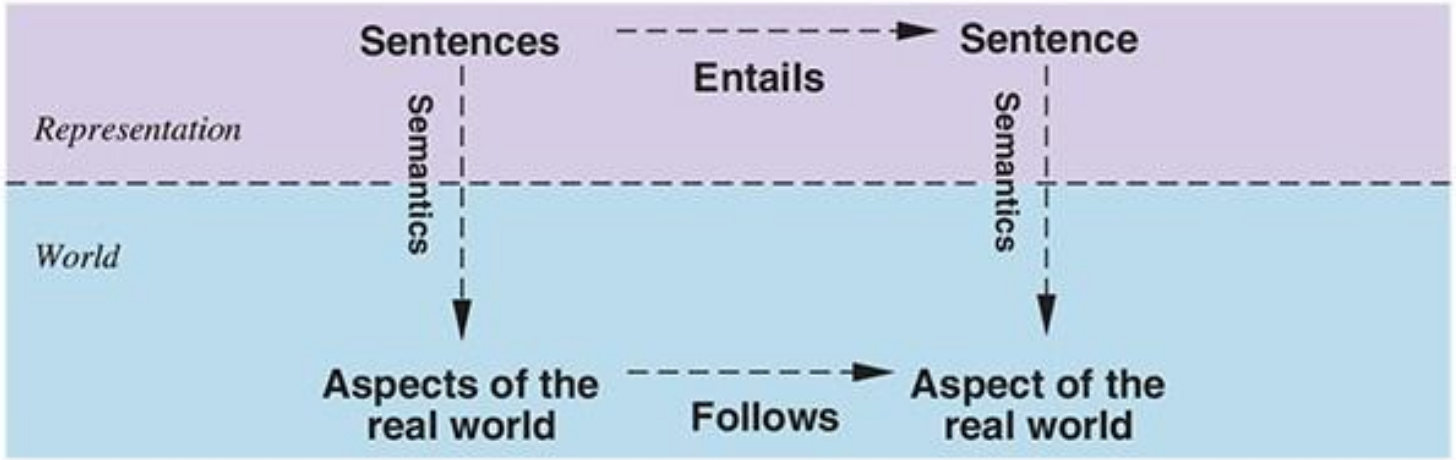
Inference: proofs

- A proof is a **demonstration** of entailment between α and β
- We use an inference algorithm to do this

- ***Sound*** (or truth-preserving) algorithm
  - everything that it claims to prove is in fact entailed
  - An inference algorithm that derives only entailed sentences
- ***Complete*** algorithm
  - everything that is entailed can be proved
  - It can derive any sentence that is entailed

Inference: proofs

"If a knowledge base KB is true in the real *world, then any sentence α derived from KB by a sound inference procedure is also true in the real world."*



- Sentences are physical configurations of the agent, and reasoning is a process of constructing new physical configurations from old ones.
- Logical reasoning should ensure that the new configurations represent aspects of the world that actually follow from the aspects that the old configurations represent.

Inference: proofs

- Method 1: ***model-checking***
  - For every possible world, if α is true make sure that is β true too
  - OK for propositional logic (finitely many worlds)
- Method 2: ***theorem-proving***
  - Search for a sequence of proof steps (applications of ***inference rules***) leading from α to β
  - e.g., from $P \wedge (P \Rightarrow Q)$, infer Q

Propositional logic syntax

- Given: a set of proposition symbols $\{X_1, X_2, \ldots, X_n\}$
  - X_i is an atomic sentence, standing for a proposition that can be True or False
- If α is a sentence then $\neg\alpha$ is a sentence (Not or Negation)
- If α and β are sentences then $\alpha \wedge \beta$ is a sentence (And or Conjunction)
- If α and β are sentences then $\alpha \vee \beta$ is a sentence (Or or Disjunction)
- If α and β are sentences then $\alpha \Rightarrow \beta$ is a sentence (Implies or Implication)
- If α and β are sentences then $\alpha \Leftrightarrow \beta$ is a sentence (If and only if)
- And p.s. there are no other sentences!

Propositional logic semantics

$$
\begin{array}{rcl}
Sentence & \rightarrow & AtomicSentence \mid ComplexSentence \\
AtomicSentence & \rightarrow & True \mid False \mid P \mid Q \mid R \mid \ldots \\
ComplexSentence & \rightarrow & (\, Sentence \,) \\
 & \mid & \neg\, Sentence \\
 & \mid & Sentence \wedge Sentence \\
 & \mid & Sentence \vee Sentence \\
 & \mid & Sentence \;\Rightarrow\; Sentence \\
 & \mid & Sentence \;\Leftrightarrow\; Sentence \\
\\
\textsc{Operator Precedence} & : & \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{array}
$$

A formal grammar of propositional logic, along with operator precedences, from highest to lowest.

Propositional logic semantics

- Semantics
  - It defines the rules for determining the truth of a sentence w.r.t a model
- Let m be a model assigning true or false to $\{X_1, X_2, \ldots, X_n\}$
- If α is a symbol then its truth value is given in m
- $\neg\alpha$ is true in m iff α is false in m
- $\alpha \wedge \beta$ is true in m iff α is true in m <u>and</u> β is true in m
- $\alpha \vee \beta$ is true in m iff α is true in m <u>or</u> β is true in m
- $\alpha \Rightarrow \beta$ is true in m unless α is true in m <u>and</u> β is false in m
- $\alpha \Leftrightarrow \beta$ is true in m iff $\alpha \Rightarrow \beta$ is true in m <u>and</u> $\beta \Rightarrow \alpha$ is true in m

Propositional logic semantics

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	*false*	*true*	*false*	*false*	*true*	*true*
false	*true*	*true*	*false*	*true*	*true*	*false*
true	*false*	*false*	*false*	*true*	*false*	*false*
true	*true*	*false*	*true*	*true*	*true*	*true*

Truth tables for the five logical connectives. To use the table to compute, for example, the value of P v Q when P is true and Q is false, first look on the left for the row where P is true and Q is false (the third row). Then look in that row under the P v Q column to see the result: true.

Some reasoning tasks

- ***Localization*** with a map and local sensing:
  - Given an initial KB, plus a sequence of percepts and actions, where am I?
- ***Mapping*** with a location sensor:
  - Given an initial KB, plus a sequence of percepts and
  - actions, what is the map?
- ***Planning***:
  - Given …, what action sequence is guaranteed to reach the goal?
- ***ALL OF THESE USE THE SAME KB AND THE SAME ALGORITHM!!***

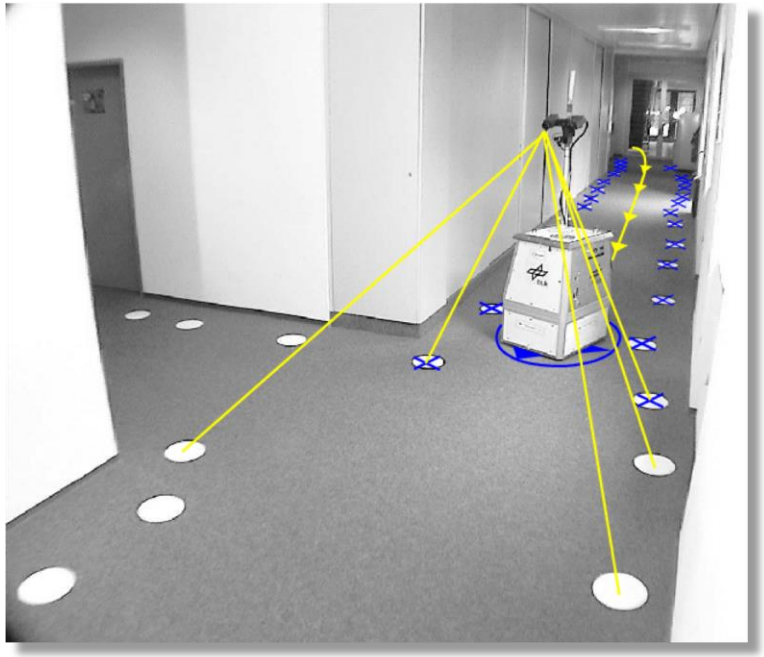


Image is from https://roboticsandautomationnews.com/2019/08/28/slam-software-paving-the-way-for-new-generation-autonomous-mobile-robots/25218/

Inference (reminder)

- Method 1: ***model-checking***
  - For every possible world, if α is true make sure that is β true too
- Method 2: ***theorem-proving***
  - Search for a sequence of proof steps (applications of ***inference rules***) leading from α to β
- ***Sound*** algorithm: everything it claims to prove is in fact entailed
- ***Complete*** algorithm: every that is entailed can be proved

Inference

- Method 2: ***theorem-proving***
  - Applying rules of inference directly to the sentences in our knowledge base to construct a proof of the desired sentence without consulting models.
- What is the advantage over Method 1: ***model-checking?***
  - If the number of models is large but the length of the proof is short, then theorem proving can be more efficient than model checking

Inference

- **Concept 1:** Logical equivalence
  - two sentences α and β are logically equivalent if they are true in the same set of models.
  - We write this as $\alpha \equiv \beta$. (Note it is different from "⇔" which is used as part of a sentence)
  - Another definition: any two sentences α and β are equivalent if and only if each of them entails the other.
    - $\alpha \equiv \beta$ if and only if $\alpha \vDash \beta$ and $\beta \vDash \alpha$
  - These equivalences play much the same role in logic as arithmetic identities do in ordinary mathematics.
- Quiz: is $P \wedge Q$ and $Q \wedge P$ are logically equivalent?

Inference

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

Inference

- Concept 2: **Validity**
  - A sentence is valid if it is true in all models.
  - Is the sentence $P \vee \neg P$ is valid?
  - Valid sentences are also known as tautologies—they are necessarily true
  - Every valid sentence is logically equivalent to True
- Deduction theorem
  - For any sentences α and β, $\alpha \vDash \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid
  - We can decide if $\alpha \vDash \beta$ by checking that $(\alpha \Rightarrow \beta)$ is true in every model
  - Or, by proving that $(\alpha \Rightarrow \beta)$ is equivalent to True

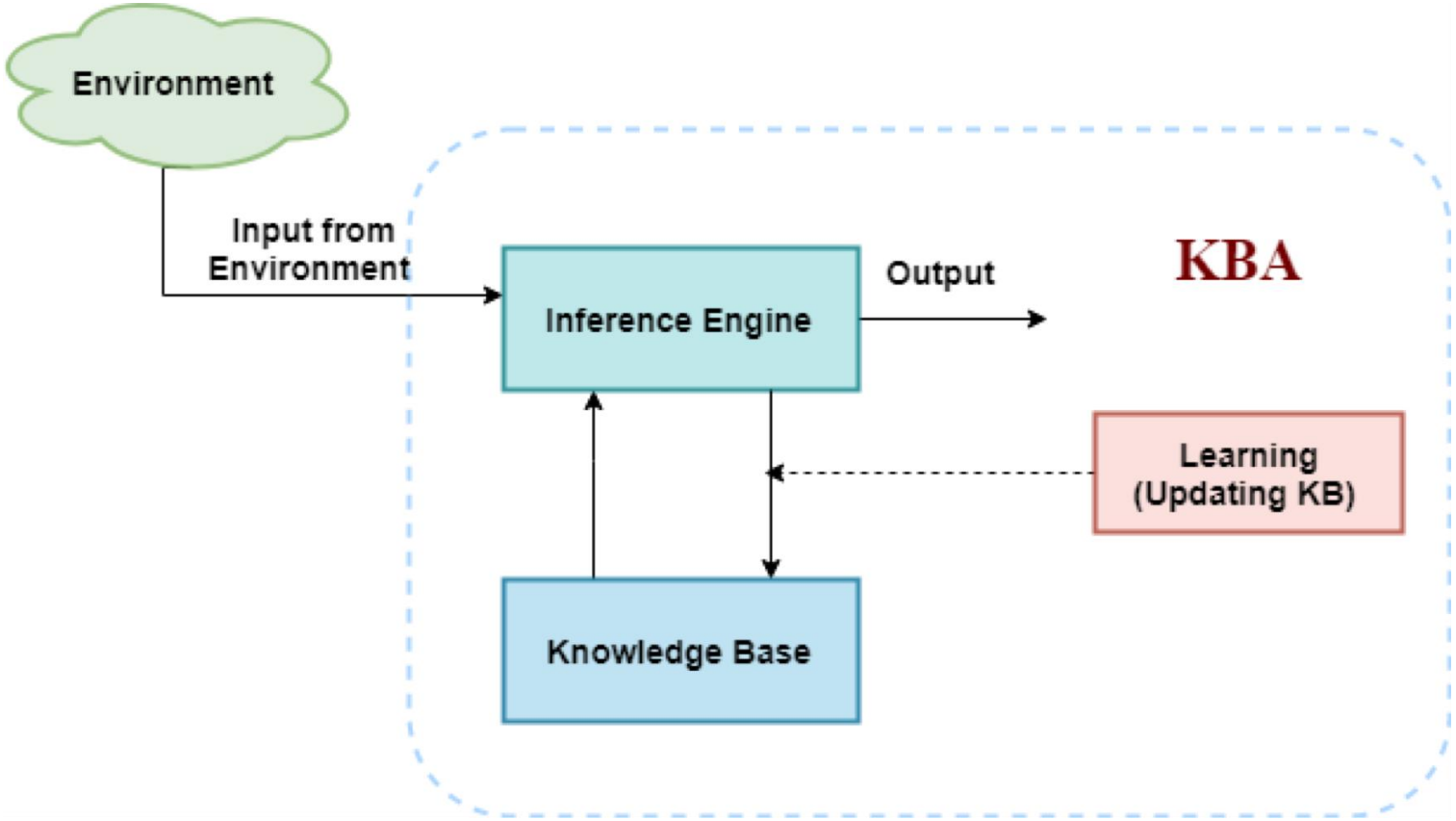
Inference

- Concept 3: **Satisfiability (SAT)**
  - A sentence is satisfiable if it is true in, or satisfied by, some model.
  - Satisfiability can be checked by enumerating the possible models until one is found that satisfies the sentence.
  - Many problems in computer science are really satisfiability problems.
- Validity and satisfiability
  - α is valid iff $\neg\alpha$ is unsatisfiable; α is satisfiable iff $\neg\alpha$ is not valid.
  - $\alpha \vDash \beta$ if and only if the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable
    - reductio ad absurdum (literally, “reduction to an absurd thing”).
    - It is also called proof by refutation or proof by contradiction.

Inference

- **Inference rule**
  - The rule that can be applied to derive a proof – a chain of conclusions that lead to the desired goal
- **Modus Ponens** *(Latin for mode that affirms)*
  - Whenever any sentences of the form **α ⇒ β** and **α** are given, then the sentence **β** can be inferred.
- **And-Elimination**
  - From a conjunction, any of the conjuncts can be inferred
    - from **α ∧ β**, infer **α**
    - from **α ∧ β**, infer **β**
- **Resolution**
  - A single inference rule used on **clauses** (disjunctions of literals), typically in **CNF** (Conjunctive Normal Form)
    - from **(A ∨ Γ)** and **(¬A ∨ Δ)**, infer **(Γ ∨ Δ)**
- **All can be used to generate sound inferences**
  - without the need for enumerating models

The architecture of a knowledge-based agent



https://www.javatpoint.com/knowledge-based-agent-in-ai

Syntax of propositional logic

1. Formula: A
2. Formula: $\neg A$
3. Formula: $\neg B \rightarrow CA$
4. Formula: $\neg A \wedge (\neg B \rightarrow C) \vee (\neg B \vee D)$
5. Formula: $\neg\neg A$
6. Formula: $A\neg B$
7. Formula: $A + B$

Which of them are valid formulas?

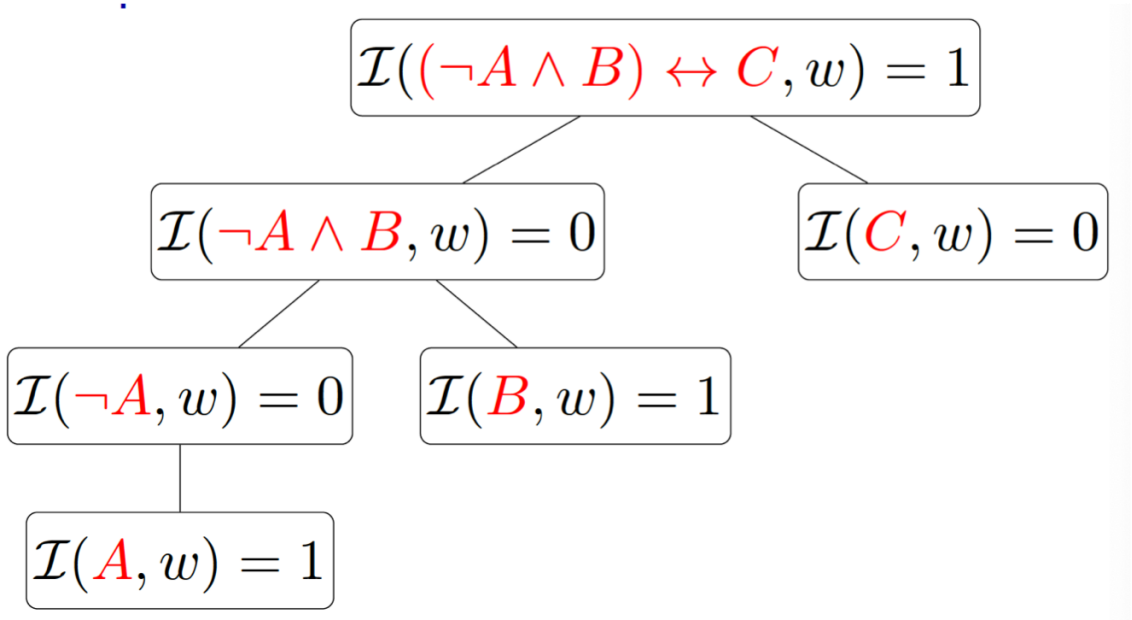
Interpretation function: example

Example: interpretation function

Formula: $f = (\neg A \wedge B) \leftrightarrow C$

Model: $w = \{A : 1, B : 1, C : 0\}$

Interpretation:



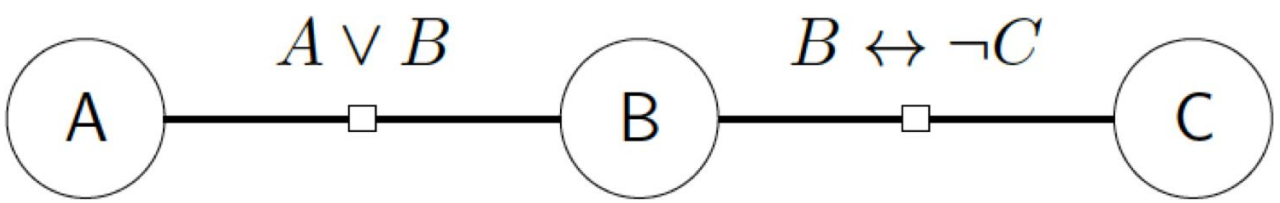
Model checking

Example: model checking

$$\text{KB} = \{A \vee B, B \leftrightarrow \neg C\}$$

Propositional symbols (variables):

$$\{A, B, C\}$$



Consistent assignment (satisfying model):

$$\{A : 1, B : 0, C : 1\}$$

Inference rules

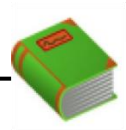
Example of making an inference:

It is raining. (Rain)

If it is raining, then it is wet. (Rain $\rightarrow$ Wet)

Therefore, it is wet. (Wet)

$$\frac{\text{Rain}, \quad \text{Rain} \rightarrow \text{Wet}}{\text{Wet}} \qquad \frac{\text{(premises)}}{\text{(conclusion)}}$$



Definition: Modus ponens inference rule

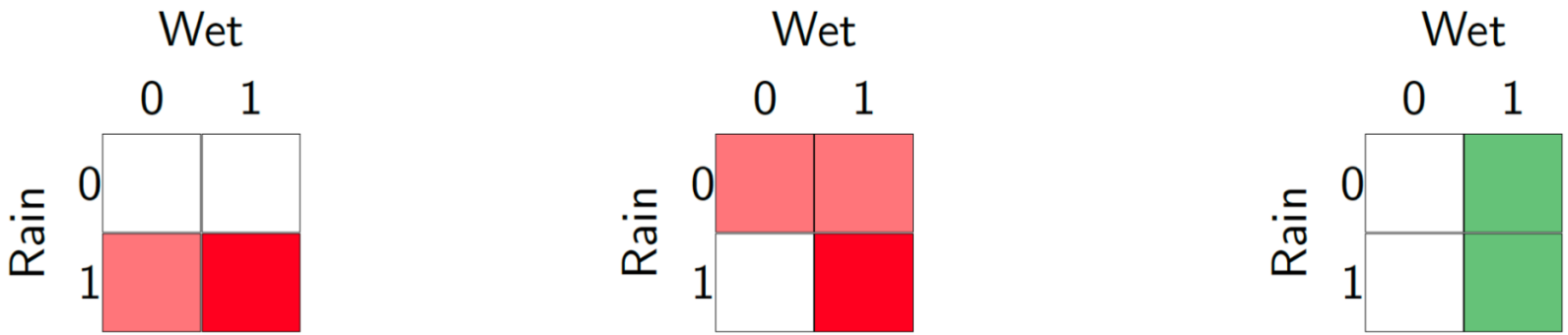
For any propositional symbols p and q

$$\frac{p, \quad p \rightarrow q}{q}$$

Soundness: example

Is $\dfrac{\text{Rain}, \quad \text{Rain} \to \text{Wet}}{\text{Wet}}$ (Modus ponens) sound?

$$\mathcal{M}(\text{Rain}) \quad \cap \quad \mathcal{M}(\text{Rain} \to \text{Wet}) \quad \subseteq ? \quad \mathcal{M}(\text{Wet})$$

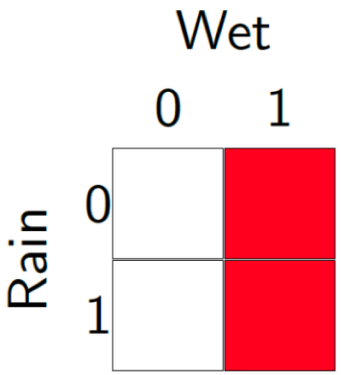


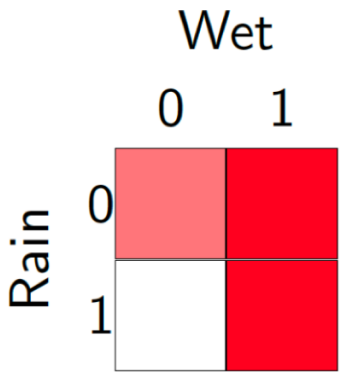
Sound!

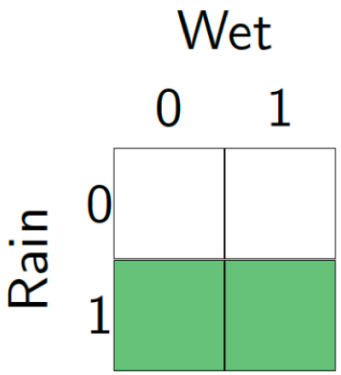
Soundness: example

Is $\dfrac{\text{Wet}, \quad \text{Rain} \to \text{Wet}}{\text{Rain}}$ sound?

$$\mathcal{M}(\text{Wet}) \quad \cap \quad \mathcal{M}(\text{Rain} \to \text{Wet}) \quad \subseteq ? \quad \mathcal{M}(\text{Rain})$$







Unsound!

Summary

- One possible agent architecture: knowledge + inference
- Logics provide a formal way to encode knowledge
  - A logic is defined by: syntax, set of possible worlds, truth condition
- A simple KB could cover the initial state, sensor model, and transition model
- Logical inference computes entailment relations among sentences, enabling a wide range of tasks to be solved
- Theorem provers apply inference rules to sentences

Acknowledgement

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Questions?

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